MIL-S-81282A
6 December 1973
SUPERSEDING
MIL-S-81282 (WP)
9 July 1965

MILITARY SPECIFICATION

STANDARD, FREE WATER DETECTOR

This specification is approved for use by all the agencies and departments of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers standard to be used with Free Water Detector conforming to MIL-D-81227.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-T-60	Tape: Packing Waterproof
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Box, Wood Cleated Plywood
PPP-B-636	Boxes, Shipping, Fibreboard

Military

MIL-D-81227	Detector Kit, Water, Automotive-Aviation Fuel
MIL-D-81248	Detector Pad, Free Water

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 6640

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitations for bid or request for proposal shall apply.

National Classification Board:

National Motor Freight Classification

(Application for copies shall be addressed to the American Trucking Associations, Inc., Attn: Tariff Order Section, 1616 P Street, N.W. Washington, D.C. 20036.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials. Materials shall conform to applicable specifications and shall be specified herein. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality suitable for the purpose intended.

3.1.1 Paper. The chemical composition of the paper shall be in accordance with type ORO T-4, as supplied by the Color Aid Company, 116 E. 27th Street, New York, N.Y. 10016 under GSA Contract GS00S 28543, or approved equivalent. The paper shall have a pressure-sensitive backing consisting of a water-insoluble adhesive with carrier or separator sheet applied over the adhesive.

3.1.2 Inks.

3.1.2.1 Formula A. Formula A ink shall consist of 100 percent Fluorescent Yellow Lithographic Ink No. 53-P-3950, as supplied by Crescent Ink Co. 464 N. 5th Street, Phila., Pa. 19123, or approved equivalent.

3.1.2.2 Formula B. Formula B shall consist of 55.6 percent (by weight) Fluorescent Yellow Lithographic Ink No. 63-P-3950 or approved equivalent, 18.5 percent (by weight) Fluorescent Rose Lithographic Ink No. 64-P-3950 or approved equivalent, 0.90 percent (by weight) of Fluorescent

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Yellow C#4, and 25.0 percent (by weight) of Dayglo Litho Signal Green 77-18 or approved equivalent. Fluorescent Yellow Lithographic Ink No. 63-P-3950 and Fluorescent Rose Lithographic Ink No. 64-P-3950 and Fluorescent Rose Lithographic Ink No. 64-P-3950 are available from Crescent Ink Co. 464 N. 5th St. Phila., Pa. 19123. Fluorescent Yellow C#4 is supplied by Patent Chemicals Inc. 355 McLean Blvd, Paterson, N.J. 07504. Dayglo Litho Signal Green 77-18 is supplied by Switzer Brothers, 4732 St. Clair Ave. Cleveland, Ohio 44103.

3.1.2.3 Formula C. Formula C ink shall consist of 100 percent of Rex Speed Job Black as supplied by Smith, Edwards, and Dunlap Co. 2867 E. Allegheny Ave. Phila., Pa. 19134.

3.1.3 Engraved plates. Duplicate plates or patterns for the engraved printing plates as manufactured by the American Photoengraving Company (66 N. Juniper St. Phila., Pa. 19107) shall be supplied by the Naval Air Propulsion Test Center, Trenton, N.J. 08628 to contractor when contract is awarded. The plates shall be designated 1, 2, and 3 (see 3.3). The contractor shall make molds of the plates, to be used in the printing process to avoid damage to the original plates.

3.1.4 Apparatus. The apparatus (6.4) shall consist of a fluorometer used to monitor the fluorescence of each step during the printing procedure.

3.2 Design. The standard shall be a comparator capable of determining the amount of free water in fuel when a detector pad, in accordance with MIL-D-81248, is compared with the applicable patch on the standard. The design of the standard shall conform to item 25 of MIL-D-81227.

3.3 Construction. The construction of the Water Detector Standard shall be in accordance with the following procedure.

Step 1 - Plate 1 (3.1.3) an art form and formula A ink (3.1.2.1) shall be used to produce a background fluorescent pattern for patches 5, 10, and 20 on the standard.

Step 2 - Plate 2 (3.1.3) a half tone and formula B ink (3.1.2.2) shall be used to produce an overall fluorescent pattern for patches 5, 10, and 20.

Step 3 - Plate 3 (3.1.3) shall be used to produce a black overprint for the visible layout of the finished standard. Steps 1, 2, and 3 shall be printed in perfect registration.

3.4 Dimensions. The trim sizes of the standard shall be 1 3/4 by 9 inch.

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3.5 Storage. Finished standards when packed in accordance with 5.3 shall be placed in a humidity controlled storage area.

3.6 Workmanship. The standard shall be so constructed and finished that it will be free, when examined visually, from defects which may affect proper functioning in service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. All sampling and inspection levels shall conform to MIL-STD-105. The inspection shall consist of examining and testing samples for tests (4.2.1) and examination of the sample of filled containers (4.2.2) to determine conformance to packaging, packing and marking requirements.

4.2.1 Examination for tests. Free water detector standards shall be subjected to the following inspections:

(a)	Examination	(4.3.1)
(b)	Registration	(4.3.2)
(c)	Density	(4.3.3)

An inspection level II and an acceptable quality level (AQL) of 1.0 shall be assigned for examination, and an inspection level of S-3 and an acceptable quality level (AQL) of 1.0 shall be assigned for testing.

4.2.2 Inspection for packaging, packing, and marking. Random samples shall be selected from each inspection lot in accordance with MIL-STD-105 at inspection level S-2 with acceptable quality level of 4.0 percent defective (see 4.3.4).

4.2.2.1 Inspection lot. An inspection lot shall consist of free Water Detector Standards that have been printed at essentially the same time under the same conditions. The Sample unit shall be one fully prepared shipping container fully prepared for delivery with the exception that it need not be sealed.

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4.3 Inspection methods.

4.3.1 Examination. When visually examined each Free Water Detector standard shall be evenly coated and free from foreign matter, holes, tears, cuts, greases, wrinkles or other imperfections which would render the standard unsuitable for its intended use. Each standard shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship and marking.

4.3.2 Registration. Each Free Water Detection Standard shall be tested visually under ultraviolet light for registration, using Free Water Detector (see 6.5).

4.3.3 Density. Each Free Water Detector Standard shall be tested for density using a fluorometer. The results shall fall between the maximum and minimum limits shown in Figure 1. These limits are defined as follows for printing steps 1 and 2 (see 3.3).

Step 1

Standard	Min. Value	Max. Value
5 ppm	6.7	9.5
10 ppm	11.0	15.8
20 ppm	16.9	29.7

Step 2

Standard	Min. Value	Max. Value
5 ppm	10.4	13.6
10 ppm	20.8	28.0
20 ppm	47.4	56.2

4.3.4 Packaging, packing and marking. Each of the fully prepared shipping containers selected as a sample unit shall be visually examined to determine that packaging, packing and marking conform to Section 5.

5. PREPARATION FOR DELIVERY

5.1 Unit container. Each standard shall be individually packaged in a light-proof, moisture-proof envelope.

5.2 Intermediate packaging. The standards, packaged as specified in 5.1, shall be further packaged in multiples of 50 in intermediate containers conforming to PPP-B-566.

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5.3 Packing.

5.3.1 Level A. Intermediate containers shall be packed in a close fitting shipping container constructed, closed and strapped in accordance with class weather-resistant compliance symbol V3C or V3S, Style RCS of PPP-B-636. Net weight of contents in each shipping container shall not exceed 65 pounds. Each fibre-board shipping container shall be further sealed with a 3 inch minimum width pressure sensitive tape conforming to Class III of PPP-T-60. The tape shall be evenly applied over the full length of all seams and closures including the manufacturer's joint and extending at least three inches onto each end and side panels as applicable. When specified, (see 6.2) for Army use, the intermediate containers shall be packed in wood-cleated plywood containers conforming to PPP-B-601, Grade A, closed in accordance with the Appendix thereto and subject to the weight limitations of PPP-B-601.

5.3.2 Level B. Intermediate containers shall be packed in a close fitting container constructed in accordance with PPP-B-636, class domestic, Style RSC. Net weight of contents in each shipping container shall not exceed 65 pounds. All flaps of the shipping container shall be securely sealed with an adhesive used commercially for sealing fiberboard boxes by application throughout the entire area of contact between the flaps or by a combination of metal stitching the bottom and sealing the top flaps with adhesive.

5.3.3 Level C. Packages that require overpacking for acceptance by the common carrier shall be packed in exterior type shipping containers in a manner that will assure safe transportation at the lowest cost to the point of delivery. Containers shall be in accordance with the Uniform Freight Classification Rules or Motor Freight Rules, as applicable.

5.4 Marking. Each container, including the unit container, shall be marked in accordance with MIL-STD-129.

5.4.1 Additional marking. Each packaging envelope shall be marked with the following:

(a) Military Specification MIL-S-81282A

(b) Standard, Free Water Detector

(c) CAUTION: Improper handling or continual exposure to sunlight, chemical fumes, etc. may cause colors of the standard to fade. User should exercise care to insure adequate protection to preclude color change.

(d) Date of printing

(e) Printer's name, address, and contract No.

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6. NOTES

6.1 Intended use. The standard covered by this specification is intended for use with Free Water Detector conforming to MIL-D-81227. The standard shall be used as a comparator to determine the degree of water contaminant in aircraft fuel when the Detector Pad (MIL-D-81248) is compared with the standard under ultraviolet light.

6.2 Ordering Data.

(a) Title and number of this specification

(b) Quantity required

(c) Level of packing required (see 5.3 and 5.3.1)

6.3 Special notice. The Free Water Detector Standards covered by this specification are not satisfactorily reproducible by the offset process. Satisfactory reproduction can be made only by using the duplicate plates or molds (3.1.3) which are made from the original plates supplied by the Naval Air Propulsion Test Center, Trenton, N.J. 08628.

6.4 The fluorometer used can be a Model III manufactured by G. K. Turner Associates 2524 Pulgas Ave. Palo Alto, California 94303 or equivalent. It has a paper chromatogram door accessory. This door scans a sample standard after each fountain adjustment to assure the correct density (see 4.3.3).

6.5 A free water detector conforming to MIL-D-81227 is required for checking the registration of the various fluorescent in printing steps.

6.6 The Naval Air Propulsion Test Center, Trenton, N.J. 08628 is the activity that is responsible for deciding whether alternate materials, that are offered, are the equivalent of those specified herein. However bidders should submit detailed information of their offerings through the procurement officer of the cognizant procuring agency for a determination as to acceptability.

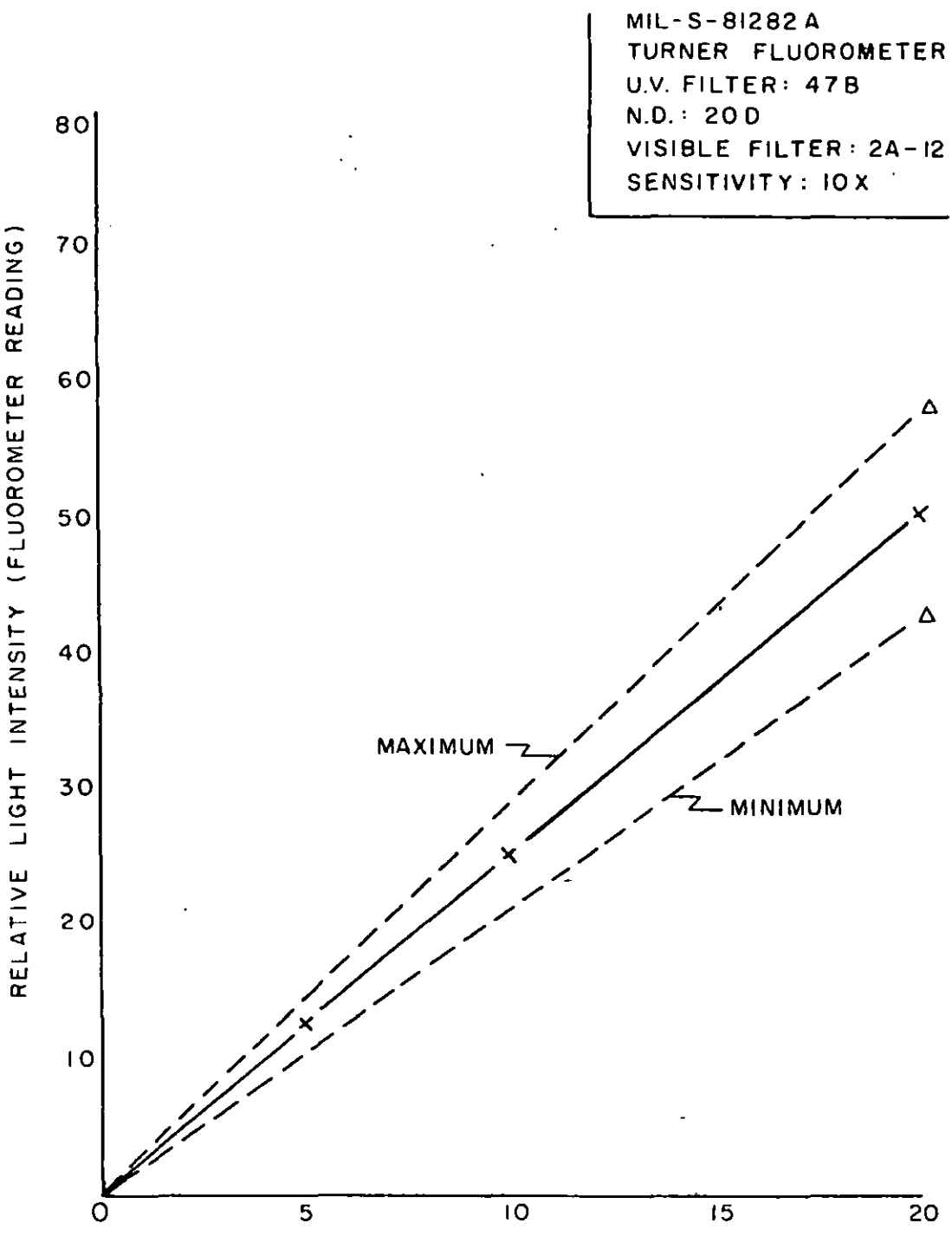
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Preparing Activity
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FREE WATER DETECTOR STANDARDS: DENSITY

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-S-81282A STANDARD, FREE WATER DETECTOR

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

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